

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		Between glomerulus and Bowman's capsule	1			1		1
		ii		Correct answer = 30 000 = 2 marks <u>15mm x 10⁶</u> (1) Evidence of Image/Actual 500		2		2	2	
	(b)	(i)		$8 - 4 - 2.6 = 1.4 \text{ kPa}$		1		1	1	
		(ii)		1. There is less protein in the {plasma/blood} (1) 2. So the osmotic pressure decreases/water potential higher/water potential less negative (1) 3. So overall pressure increases and {more filtrate is produced/rate is faster}(1)		3		3		
		(iii)		Dilate the afferent vessel/afferent vessel gets wider/ muscle (in walls) relaxes (1) Constrict the efferent vessel/efferent gets narrower/ muscles (in walls) contract (1)			2	2		
	(c)			They are too large to {diffuse back in/pass through/to be reabsorbed}(1) There are no {specific carriers/transport proteins/carrier proteins/channel proteins} for them (1)			2	2		

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	(d)	(i)		(Only) water reabsorbed/water leaves tubule by osmosis/diffusion (1)		1		1		
		(ii)		point drawn at 40 au/100% (1) Line drawn with ruler between points (1) No extrapolation/sketchy line		2		2	2	
	(e)	(i)		{Provide/Release} energy/{produce/provide/release/synthesise} ATP (1) Reject produce/make energy For active {transport/uptake}/for pumping correct named molecule (1)	2			2		
		(ii)		{Give larger/increased/large} surface area <u>for absorption/or description of</u>	1			1		
	(f)			Any 4 × (1) from: 1. Ammonia is <u>more</u> toxic than urea (1) 2. so needs to be diluted/washed away constantly/more soluble than urea (1) 3. Mud: Need to conserve water(1) 4. Urea needs less water for excretion/if ammonia was used, a lot of water would be lost (1) 5. Urea can be stored (because it is less toxic) (1)		2	2	4		
				Question 7 total	4	11	6	21	5	0